

Colon Cancer Survival Prediction

► Using Machine Learning Models

Presented by

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Data Sources

Data Source #	Name/ Short Description	Source URL	Type: - API - Web page - File	List of Fields	Format: - json - xml - csv - sql - other	Have tried to access/collect data with python? yes/no	Estimated data size, number of data points you plan to use
1	NIH Colorectal Cancer Clinical Trials	https://clinicaltrialsapi.cancer.gov/api/v2	API	Healthcare	json	Yes	300 or more
2	Colorectal Cancer Global Dataset	https://www.kaggle.com/datasets/ankushpanday2/colorectal-cancer-global-dataset-and-predictions/discussion	File	Healthcare	CSV	Yes	300 or more
3	Colorectal Cancer Dietary and Lifestyle Dataset – colon cancer risk factors	https://www.kaggle.com/datasets/ziya07/colorectal-cancer-dietary-and-lifestyle-dataset	File	Healthcare	CSV	Yes	300 or more
4	CDC Wonder	https://wonder.cdc.gov/wonder/help/wonder-api.html	Web	Healthcare	XML	Yes	300 or more

Introduction:

This project explores how machine learning can be used to understand and predict colon cancer survival outcomes across different demographics. Colon cancer is increasingly being diagnosed in younger populations, challenging the long-held assumption that it primarily impacts older adults and making predictive modeling more critical than ever. By leveraging clinical trial intervention data from the NIH Cancer Clinical Trials API, mortality and survival statistics from CDC WONDER, a global colorectal cancer incidence dataset, and lifestyle risk factor dataset. This project aims to build models capable of predicting survival rates across age groups, genders, and countries and external factors. The goal is to demonstrate how combining clinical, epidemiological, and lifestyle data can uncover patterns that support earlier, more targeted interventions in colon cancer care.

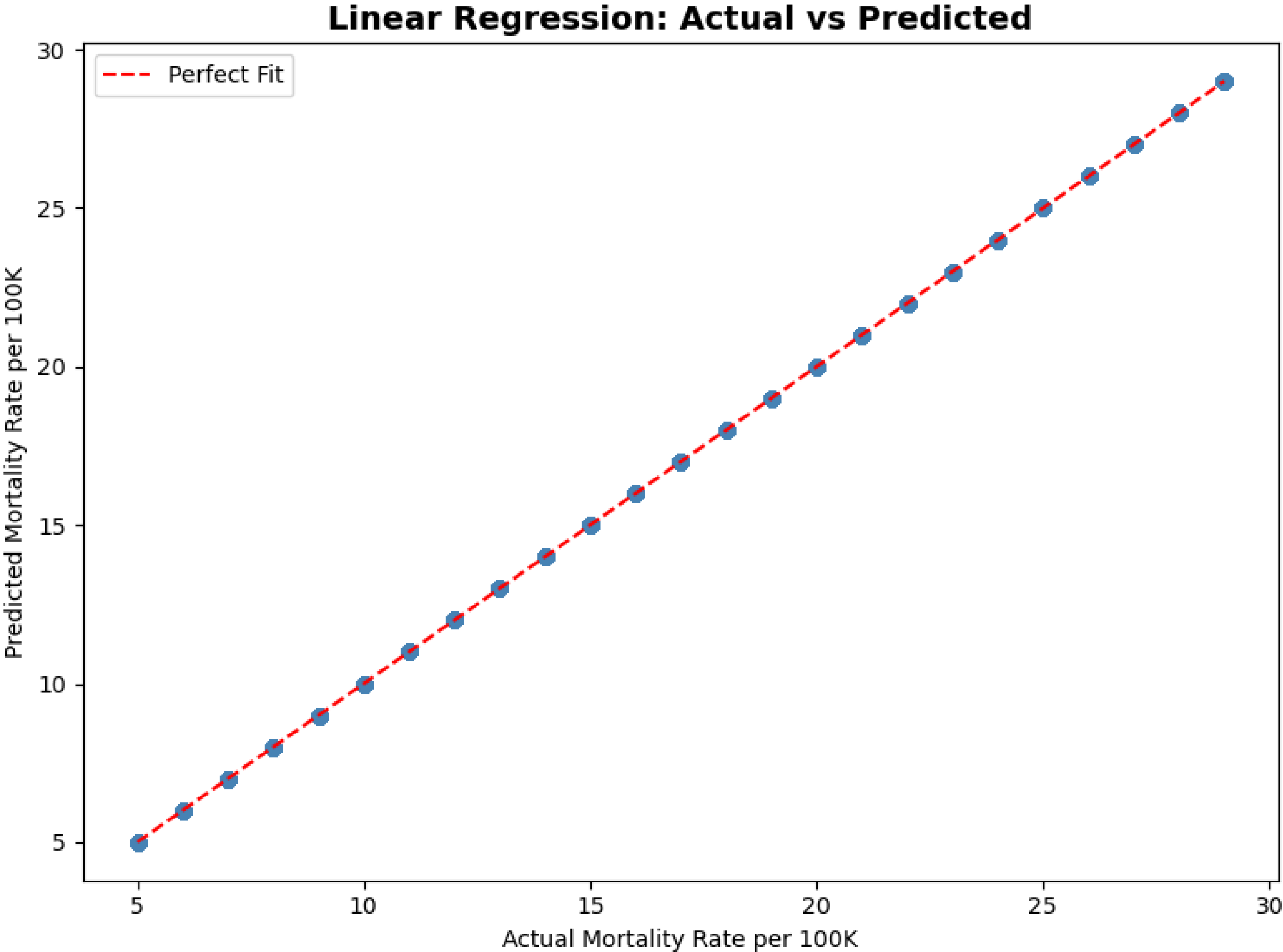


Results Summary

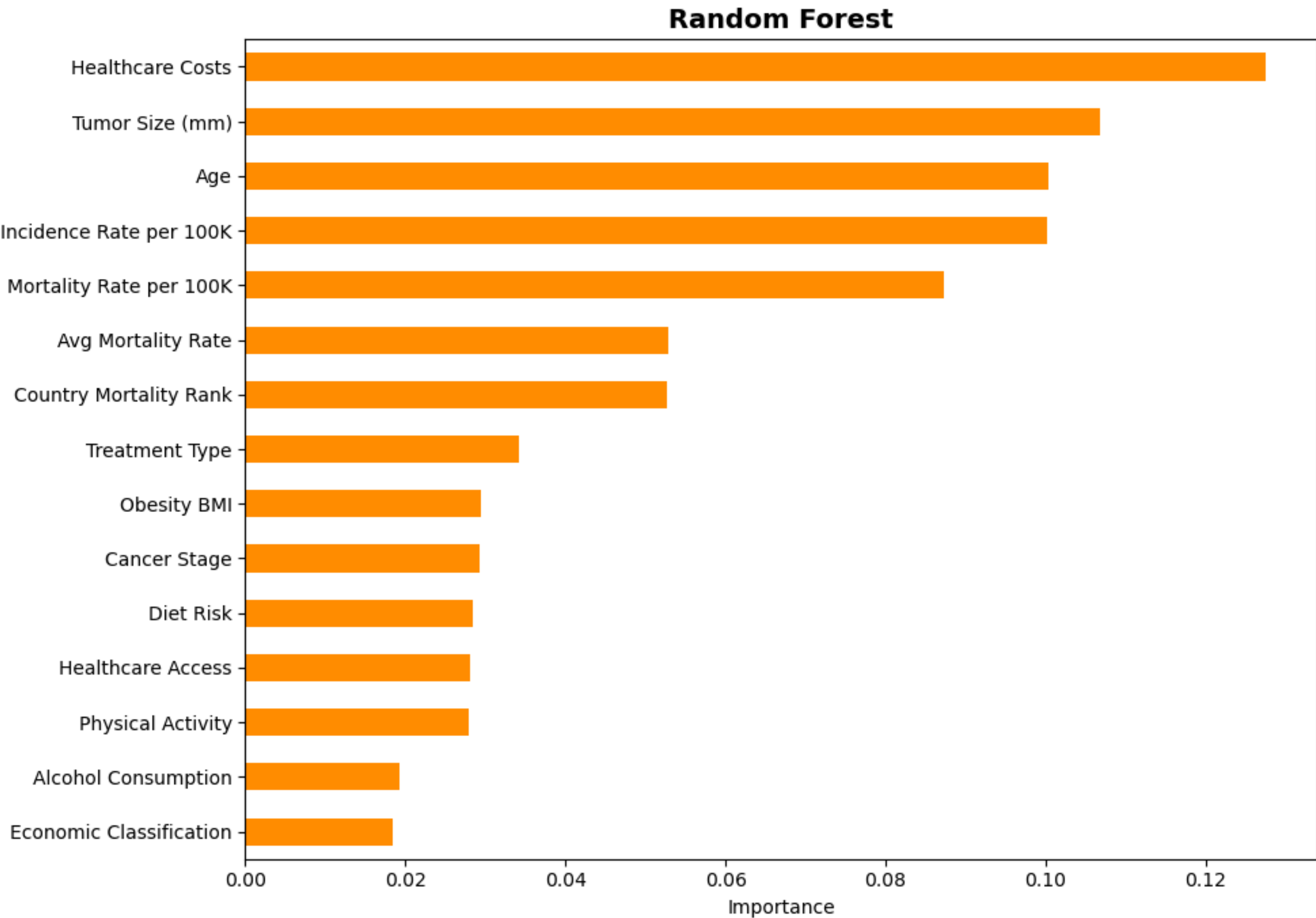
- The linear regression model performed strongly, with predicted mortality rates closely matching actual values. The Random Forest model identified healthcare costs, tumor size, and age as the top predictors of survival, while lifestyle factors ranked lower in importance. Among the NIH clinical trials, immunotherapy had the highest number of trials with survival data (154), followed by chemotherapy (119) and targeted therapy (105). The classification model performed better at predicting survival (19,056 correct) than mortality (723 correct), reflecting a common challenge with imbalanced datasets. Lastly, country mortality rankings showed a near uniform average of approximately 17 per 100K across all 16 countries, highlighting colon cancer as a global burden.



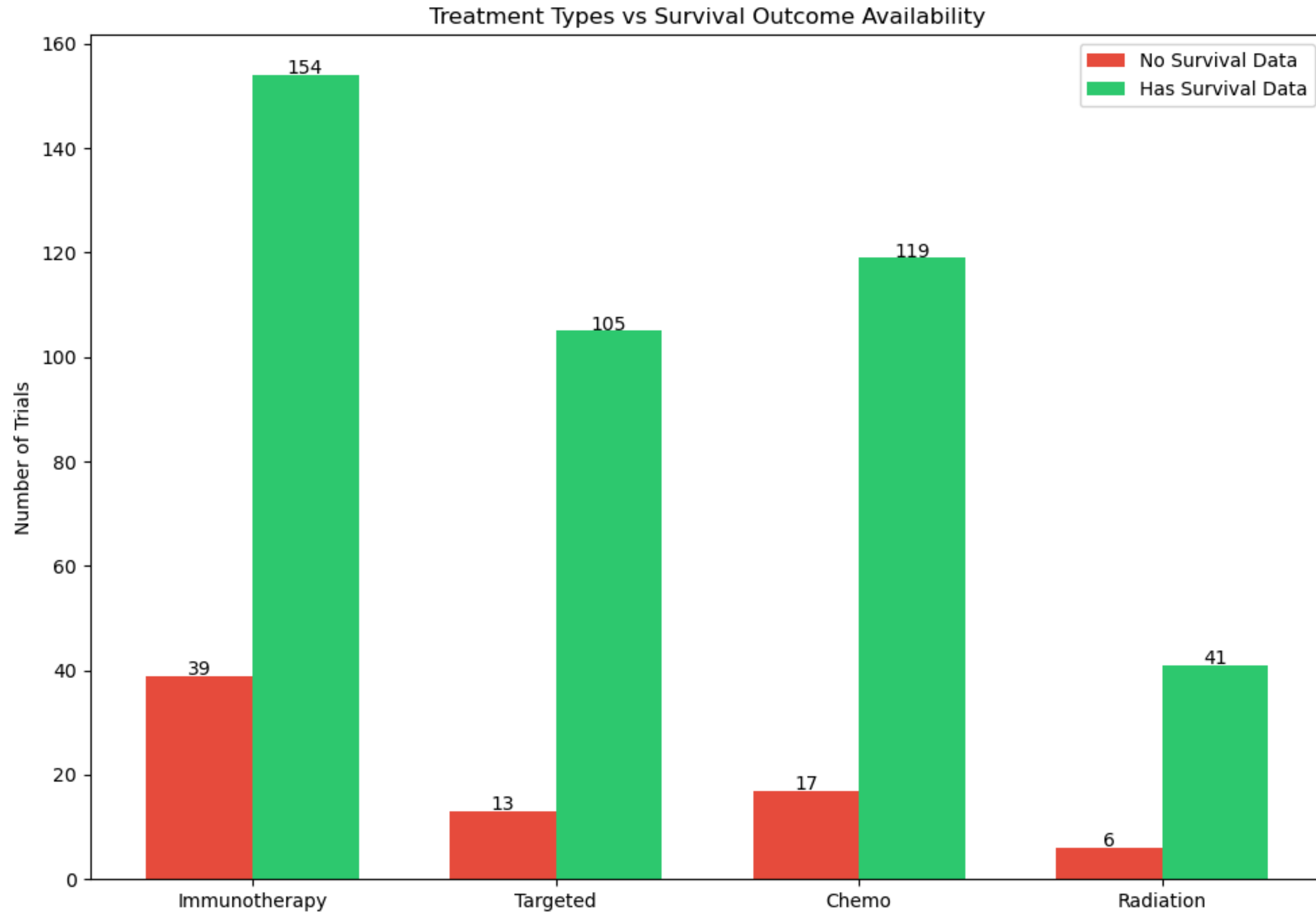
Model Results : Linear Regression Country Mortality Rankings



Model Results : Random Forest Classifier



Model Results: Bar Graph



Project Challenges

Despite successfully obtaining an API key and constructing properly authenticated requests, multiple endpoints returned inconclusive data with little relevance to the project's end goals, requiring a shift to an alternative API data source. More broadly, healthcare datasets are inherently difficult to work with due to HIPAA regulations, which restrict access to individual level patient data and limit publicly available information to aggregated statistics. Medical datasets are also often fragmented, inconsistently formatted, and subject to suppressed data points, all of which complicate preprocessing and model training. Working with large CSV files further presented challenges, as processing and merging high-volume records required careful memory management and cleaning before any meaningful analysis could be performed. Class imbalance also proved challenging, as survival cases far outnumbered death records, making it difficult for models to accurately predict mortality outcomes.



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Questions ?

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**Thank You for
Your
Attention !**